

Math 421/510: Homework # 4

1. Let X be a reflexive (and separable) normed vector space, and let $\{x_n\} \subset X$ be a bounded sequence ($\|x_n\| \leq R$ for all n , for some $R > 0$):
 - (a) show there is a subsequence x_{n_k} converging weakly to some $x \in X$;
 - (b) show that $\|x\| \leq \liminf_{k \rightarrow \infty} \|x_{n_k}\|$ (*hint: Q5 of HM 3 may help*);
 - (c) if X is Hilbert, show that $x_{n_k} \rightarrow x$ strongly (in norm) $\iff \lim_{k \rightarrow \infty} \|x_{n_k}\| = \|x\|$
(hint: express $\|x_{n_k} - x\|^2$ using the inner product).

Solution.

- (a) First, we use the hint given by Prof. Yavicoli that if a space's dual is separable then the space is separable. This comes from exercise 25 in section 5.2, that says that if $\mathcal{X}$ is a Banach space, and $\mathcal{X}^*$ is separable, then $\mathcal{X}$ is separable.

Let $Q \subset \mathcal{X}$ be a countable dense subset of $\mathcal{X}$. Then, since $\mathcal{X}$ is reflexive, we have that $\mathcal{X} \cong \mathcal{X}^{**}$. Then, the map $x \rightarrow \hat{x}$ is an isometry, so $\hat{\cdot}$ maps Q into a countable dense subset of $\mathcal{X}^{**}$, so $\mathcal{X}^{**}$ is separable. Now, since $\mathcal{X}^*$ is a set of linear functionals, it is the set of linear maps into a complete vector space (that is, $\mathbb{C}$), this implies $\mathcal{X}^*$ is complete by proposition 5.4 in Folland, so it is a Banach space.

Then, we were given by Prof. Yavicoli that we can use Exercise 25 in section 5.2 in Folland without proof, that if X is Banach and $\mathcal{X}^*$ is separable then $\mathcal{X}$ is separable. In particular, $(\mathcal{X}^*)^*$ separable and $\mathcal{X}^*$ Banach implies that $\mathcal{X}^*$ separable.

Then, we can apply Banach-Alaoglu, which is theorem 4.48 in the notes, that says (applied to this context) that $\mathcal{X}^*$ separable implies that any bounded sequence in $(\mathcal{X}^*)^*$ has a weak* convergent subsequence. In particular, this means that a sequence $\hat{x}_n$ in $\mathcal{X}^{**}$ has a subsequence $\hat{x}_{n_k}$ converging to some $\hat{x} \in \mathcal{X}^{**}$ with respect to weak* convergence. However, this means precisely for all $f \in X^*$, $\hat{x}_{n_k}(f) \rightarrow \hat{x}(f)$, which means that $f(x_{n_k}) \rightarrow f(x)$. This, then, is the same statement as $x_{n_k} \rightarrow x$ with respect to weak convergence in $\mathcal{X}$. Therefore, x_{n_k} is a subsequence of x_n that converges to $x \in \mathcal{X}$ weakly, as desired.

- (b) First, since $\|x_{n_k}\|$ is bounded below (by 0), there exists some subsequence $\{x_{n_j}\} \subseteq \{x_{n_k}\}$ such that $\lim_{j \rightarrow \infty} \|x_{n_j}\| = \liminf_{k \rightarrow \infty} \|x_{n_k}\|$. Since x_{n_j} is a subsequence of x_{n_k} , $x_{n_j} \rightarrow x$. Now, since $f \in \mathcal{X}^*$ are linear functionals, we have that for $\alpha \neq 0, \alpha \in \mathbb{C}$, $f(\alpha x_{n_j}) \rightarrow f(\alpha x) \iff f(x_{n_j}) \rightarrow f(x)$. Now, suppose $R > \liminf_{k \rightarrow \infty} \|x_{n_k}\|$. Then, there exists some N such that for $j > N$, $\|x_{n_j}\| < R$. This then implies that, for such j , $\|x_{n_j}\|/R \leq 1$, so in the notation of Homework 3 question 5, $x_{n_j}/R \in B$ where $B = \{x \in \mathcal{X} : \|x\| \leq 1\}$. Then, since for all $f \in X^*$ $f(x_{n_j}/R) \rightarrow f(x/R)$ (x_{n_j}/R converges weakly to x/R), and B is closed under weak convergence by Homework 3 question 5, we have that $x/R \in B$. However, this implies $\|x/R\| \leq 1$, so $\|x\| \leq R$. Thus, for all $R > \liminf_{k \rightarrow \infty} \|x_{n_k}\|$, $\|x\| \leq R$, so $\|x\| \leq \liminf_{k \rightarrow \infty} \|x_{n_k}\|$.
- (c) We have that $\|x_{n_k} - x\|^2 = \|x_{n_k}\|^2 - 2\operatorname{Re} \langle x_{n_k}, x \rangle + \|x\|^2$. However, since $\langle \cdot, x \rangle$ is a linear functional, we have that $\langle x_{n_k}, x \rangle \rightarrow \langle x, x \rangle = \|x\|^2$ since $x_{n_k} \rightarrow x$ weakly.

Therefore, $\|x_{n_k} - x\|^2 \rightarrow \|x_{n_k}\|^2 - \|x\|^2$, and we have that this is greater or equal to 0 by part b). Thus, this limit lets us conclude that $\|x_{n_k} - x\|^2 \rightarrow 0$ iff $\|x_{n_k}\| \rightarrow \|x\|$, as desired.

2. Show that $C_0(\mathbb{R}^n)$ is the closure of $C_c(\mathbb{R}^n)$ in the supremum norm $\|\cdot\|_\infty$.

Solution.

We need to show two things. First, if $\{f_n\} \subset C_c(\mathbb{R}^n)$ is such that $f_n \rightarrow_{\|\cdot\|_\infty} f$ for $f \in L^\infty(\mathbb{R}^n)$, then $f \in C_0(\mathbb{R}^n)$. Next, if $f \in C_c(\mathbb{R}^n)$, then there exists some sequence in $C_0(\mathbb{R}^n)$ that converges to f .

($\rightarrow$) Let $\{f_n\} \subset C_c(\mathbb{R}^n)$. We start by showing that $C_0(\mathbb{R}^n)$ is a complete subspace of $L^\infty(\mathbb{R}^n)$ with respect to the uniform norm $\|\cdot\|_\infty$. First, we note that $C_0(\mathbb{R}^n)$ is bounded, because on a compact domain any continuous function is bounded and outside the compact domain we have $|f(x)| \rightarrow 0$ as $\|x\|_2 \rightarrow \infty$. Next, we also have $|f(x) + \lambda g(x)| \rightarrow 0$ for $\|x\|_2 \rightarrow 0$ for $f, g \in C_0(\mathbb{R}^n)$. Finally, we proved in Math 321 that the space of complex valued, bounded, continuous functions is complete with respect to the uniform norm. Thus, if $\{g_n\} \subset C_0(\mathbb{R}^n)$ is a Cauchy sequence, then it converges to some $g \in C(\mathbb{R}^n)$ w.r.t. the uniform norm, then there exists some N such that $\|g_n - g\|_\infty < \varepsilon/2$. But then, for a fixed $n > N$ there exists some R such that for $\|x\|_2 > R$, we have $g_n(x) < \varepsilon/2$. Thus, for $\|x\|_2 > R$,

$$|g(x)| \leq |g(x) - g_n(x)| + |g_n(x)| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Therefore, $g \in C_0(\mathbb{R}^n)$, so $C_0(\mathbb{R}^n)$ is a closed subset of a complete set, so it is complete.

Now, if $f_n \rightarrow f$ with $f \in L^\infty(\mathbb{R}^n)$, then f_n is Cauchy, and since $C_c(\mathbb{R}^n) \subset C_0(\mathbb{R}^n)$ and $C_0(\mathbb{R}^n)$ is complete, we have that f_n converges to some $h \in C_0$. But since $h \in L^\infty$, we have that $f = h \in C_0(\mathbb{R}^n)$. Therefore, every sequence in $C_c(\mathbb{R}^n)$ that converges to some $f \in L^\infty(\mathbb{R}^n)$ will then have $f \in C_0(\mathbb{R}^n)$.

($\leftarrow$) Suppose $f \in C_0(\mathbb{R}^n)$. We show that there exists a sequence f_n in $C_c(\mathbb{R}^n)$ such that $f_n \rightarrow f$. Let

$$f_n(x) = \begin{cases} f(x) & \text{if } \|x\|_2 < n \\ f(x \frac{n}{\|x\|_2}) \cdot (n+1 - \|x\|_2) & \text{if } n \leq \|x\|_2 < n+1 \\ 0 & \text{otherwise} \end{cases}.$$

In particular, $\text{supp}(f_n) \subset \|x\|_2 \leq n+1$, so $f_n \in C_c$. f_n is continuous inside $\|x\|_2 < n$ because f is continuous. It is continuous at the boundary $\|x\|_2 = n$ because then $xn/\|x\|_2 = x$ so $f(x) = f(xn/\|x\|_2) = f(x \cdot 1)$. It is continuous in the annulus $n < \|x\|_2 < n+1$ because it is the composition and product of functions ($\|x\|_2$ is continuous and doesn't take 0 in this annulus). Finally, it is continuous at the boundary $\|x\|_2 = n+1$ because

$$f(x \frac{n}{\|x\|_2})(n+1 - \|x\|_2) = f(x \frac{n}{\|x\|_2})(n+1 - n+1) = 0.$$

Next, $|f_n(x) - f(x)| = 0$ when $\|x\|_2 \leq 0$, and $|f_n(x) - f(x)| \leq |f(xn/\|x\|_2)(1) - f(x)| \leq \sup_{\|x\|_2 \geq n} |f(x)|$, and $\leq \sup_{\|x\|_2 \geq n+1} |f(x)|$ for $\|x\|_2 \geq n+1$. Therefore, in every case we can see that we always have $|f_n(x) - f| \leq 2 \sup_{\|x\|_2 \geq n} |f(x)|$, which goes to zero as $n \rightarrow \infty$ since $f \in C_0(\mathbb{R}^n)$. Therefore, $f_n \rightarrow f$, as desired.

Thus, $\overline{C_c(\mathbb{R}^n)} = C_0(\mathbb{R}^n)$.

3. The solution of the initial value problem for the heat equation on $\mathbb{R}^n$:

$$\begin{cases} \partial_t u = \Delta u, & (x, t) \in \mathbb{R}^n \times (0, \infty), \\ u(x, 0) = f(x), \end{cases}$$

is given by

$$u(x, t) = (4\pi t)^{-n/2} \int_{\mathbb{R}^n} e^{-\frac{|x-y|^2}{4t}} f(y) dy.$$

(a) by expressing $u(x, t)$ as convolution with an approximate identity, show that if $f \in L^p$ ($1 \leq p < \infty$), then

$$\lim_{t \rightarrow 0^+} \|u(\cdot, t) - f\|_p = 0;$$

(b) give an example to show that we may not have $\lim_{t \rightarrow 0^+} u(x, t) = f(x)$ pointwise.

Solution.

(a) First, we check that $\phi_t(x) = (4\pi t)^{-n/2} e^{-\frac{|x|^2}{4t}}$ is an approximate identity. Let t be arbitrary. Then, we recognize the Gaussian integral, and by Fubini-Tonelli so that we have

$$\begin{aligned} \int_{\mathbb{R}^n} \phi_t(x) dx &= \int_{\mathbb{R}^n} (4\pi t)^{-n/2} e^{-\frac{|x|^2}{4t}} \\ &= \left((4\pi t)^{-1/2} \int_{\mathbb{R}} e^{-x^2/4t} dx \right)^n \\ &= (1)^n \text{ (Gaussian Integral)} \\ &= 1. \end{aligned}$$

Then, we see that $u(x, t) = \phi_t * f = f * \phi_t$. However, then, by Theorem 8.14 a, we have that if $f \in L^p$, then $f * \phi_t \rightarrow af$ in the L^p norm as $t \rightarrow 0$. In particular, since ϕ_t is an approximate identity, $a = \int \phi_t = 1$. However, the statement $f * \phi_t \rightarrow f$ in the L^p norm is precisely

$$\lim_{t \rightarrow 0^+} \|u(\cdot, t) - f\|_p = \lim_{t \rightarrow 0^+} \|f * \phi_t - f\|_p = 0.$$

(b) Consider

$$f(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ -1 & \text{if } x \in [-1, 0) \\ 0 & \text{else} \end{cases}$$

Then,

$$u(0, t) = \int_{-1}^0 \phi_t(-y) dy - \int_0^1 \phi_t(-y) dy.$$

However, since $\phi_t(x)$ is a symmetric function, and both of these integrals are finite, we have $u(0, t) = 0$. Thus, $\lim_{t \rightarrow 0^+} u(0, t) = 0 \neq f(0)$.